Introduction to Inference, Pt 2 (Revised)

CSU, Chico Math 351

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Hypothesis Testing Framework

The framework of dividing experimental conclusions into two possibilities is known as **hypothesis testing**.

- Establish hypothesis: null (H_0) and alternative (H_1).
  - H_0 and H_1 are statements about population parameters
- Collect data / define relevant parameters (such as α).
- Analyze data (calculate summary statistics) – we'll most often be estimating μ – and **p-value**.
- Make conclusion by comparing p-value to **level of significance** α.

level of significance

We define a largest level at which we are willing to incorrectly conclude. We call this value the level of significance, and give it the symbol α[1].

level of significance

The largest probability of incorrectly rejecting H_0 when in fact H_0 is true.

[1] Common values of α are 0.05 and 0.01, but these are not the only possibilities.

Step 1: Define hypotheses

The null and alternative hypotheses generally follow some conventions.

- H_0 declares the parameter of interest to be **equal** to some specific value μ_0 (or p_0 for proportions).
  - $H_0 : \mu = \mu_0$
- H_1 declares the (same) parameter of interest to be less than, greater than, or not equal to the same value in the null hypothesis H_0 – the researcher chooses one **before** conducting the test. In other words, an entire range of values rather than a single value.
  - $H_1 : \mu \ \{<, >, \neq\} \ \mu_0$.

Step 2: Collect data

Define all values needed for your hypothesis test.

- Null hypothesis value (μ_0 or p_0)
- Sample statistic ($\bar{X}$ or $\hat{p}$)
- Sample size (n)
- Sample error ($\frac{\sigma}{\sqrt{n}}$ or $\sqrt{\frac{p_0 * (1 - p_0)}{n}}$)
- Significance level α

Step 3: Analyze data

From the Central Limit Theorem, we know that $\bar{X} \sim N(\mu_0, \frac{\sigma^2}{n})$. Therefore, the test statistic z, obtained by **standardizing** $\bar{X}$, follows a standard Normal distribution:

$$z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$$

For proportions, we instead have:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 * (1 - p_0)}{n}}} \sim N(0, 1)$$

z represents, roughly, how many standard deviations away our **observed** sample statistic is from the **expected** null hypothesis value.

Step 3: Analyze data (p-value definition)

Given z, we can calculate its corresponding **p-value**, which helps us decide between H_0 and H_1.

p-value

The probability of observing the test statistic we did, or something more extreme[2], assuming the null hypothesis is true.

This is represented by the area under curve of the standard Normal distribution for values of z **more extreme** than what we observed, which we find using the cumulative distribution function (`pnorm()` in RStudio).

[2] smaller, larger, or both, depending on the alternative hypothesis.

Step 3: Analyze data (calculating p-value)

What "more extreme" means depends on the symbol used in the alternative hypothesis.
For a two-sided hypothesis test ($H_1 : \mu \neq \mu_0$) the p-value is equal to:

```
2*(1 - pnorm(abs(z)))
```

Since N(0,1) is symmetric, This gives the area under the curve for values larger than $|z|$ or less than $-|z|$, and works for both positive and negative z.

Step 4: Conclusions

A small p-value (less than α) implies a z this extreme is unlikely under the null hypothesis, and it should be rejected.
However, a p-value larger than α implies such an extreme value could have simply happened by random chance, and we cannot disprove the null hypothesis. In short:

$p - value < \alpha$: REJECT H_0
$p - value > \alpha$: FAIL TO REJECT H_0

Since H_0 is a specific value rather than a range of values, we don't ever "accept" the null hypothesis–it could still be a different value. We merely lack the evidence to disprove it.

Hypothesis Test, example

Suppose a tech company wants to test a novel web browser. They sample 1014 randomly selected websites and make a simple decision: this website was displayed properly on the browser or it was not. They found that 984 websites displayed correctly. Test whether or not the browser displays 99% of web pages correctly, and compare your conclusion to a confidence interval. Choose $\alpha = 0.05$.

- Establish hypotheses: H_0 and H_1.
- Collect data.
- Analyze data $\Rightarrow$ calculate summary statistic and p-value.
- Make conclusion $\Rightarrow$ decide H_0 or H_1 via p-value ? α

Hypothesis Test, another example

Hypothesis Test:

$$H_0 : p = .99 \quad \text{versus} \quad H_1 : p \neq .99$$

NOTE: When using a hypothesis test, the value of p in the null hypothesis should be used to calculate sigma.

```r
n <- 1014
p <- 0.99 # Population parameter according to H_0
phat <- 984/n # Estimate for the population parameter
sigma <- sqrt(p*(1-p)) # St. dev for Bernoulli variable
z <- (phat - p)/(sigma/sqrt(n)) # assume H_0 true
2*(1 - pnorm(abs(z))) # compare to alpha = 0.05

## [1] 1.826257e-10
```

Hypothesis Test, example

$100(1 - 0.05)\%$ Confidence interval.

NOTE: A confidence interval doesn't have a value of p to reference when calculating sigma. Use $\hat{p}$ instead.

```
n <- 1014
phat <- 984/n
alpha <- 0.05
sigma <- sqrt(phat*(1-phat))
se <- sigma/sqrt(n)
z.star <- qnorm(1-alpha/2)
ME <- se * z.star
c(phat - ME , phat + ME)

## [1] 0.9599850 0.9808434
```

Comparing the hypothesis test and confidence interval

Since the p-value is less than our level of significance $\alpha = 0.05$, we **reject** H_0, concluding that a proportion of pages other than 0.99 are displayed correctly.

The equivalent (95%) 2-sided confidence interval did **not** include the null hypothesis value of $p = 0.99$. This is in line with our rejection of the null hypothesis in our hypothesis test.

What about one-sided hypothesis tests?

The hypothesis test above may have been better designed as a one-sided hypothesis test, with $H_1 : p < .99$.

However, a two-sided confidence interval can't be used to describe a one-sided hypothesis test–we will instead need to consider a **one-sided confidence interval** to make a direct comparison.

We'll see examples of how to set these up in the next section.

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Hypothesis Test, example

Consider a data set consisting of number of users connected to a server on a randomly chosen day. Measurements were taken every minute for 100 minutes starting at a randomly chosen time. The data indicated a sample mean of $\bar{X} = 137$ users with a population standard deviation of 40. The company that owns this server claims that on any given day 150 users are connected to its server, but you believe that claim is **not** true[3]. Test this claim at the significance level of $\alpha = 0.05$.

- State hypotheses and choose level of significance.
- Collect data.
- Calculate summary statistics and p-value.
- Conclude.

[3] A more realistic hypothesis test for the company might attempt to claim that more than 150 customers connect on any given day – for marketing purposes.

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When α **matches** between the confidence interval and hypothesis test, and the hypothesis test has a **two-sided alternative**, we can safely compare the confidence interval to the hypothesis test.

- Did the confidence interval contain the null value?
- Which hypothesis then seems more likely given the confidence interval?

Confidence Intervals related to Hypothesis Tests, be careful

This relationship between confidence intervals and hypothesis tests is tricky when the alternative hypothesis is **one-sided** ($H_1 : \mu < \mu_0$ or $H_1 : \mu > \mu_0$). If you have a two-sided confidence interval, it will not accurately represent a one-sided hypothesis test. However, you can still use a **one-sided** confidence interval for these cases.

One-Sided Confidence Intervals

One-sided confidence intervals will have a different construction depending on the equivalent hypothesis test:

If testing hypothesis tests with the alternative hypothesis $H_1 : \mu > \mu_0$, use an **upper** confidence interval:[4]

$$[\bar{X} - z^*_{1-\alpha} \cdot \sigma_{\bar{X}}, \infty).$$

If testing hypothesis tests with the alternative hypothesis $H_1 : \mu < \mu_0$, use a **lower** confidence interval:

$$(-\infty, \bar{X} + z^*_{1-\alpha} \cdot \sigma_{\bar{X}}].$$

Note that these confidence intervals will extend to infinity in one direction, yet they will still contain the same area under the curve expected of a 100(1-α)% confidence interval.

[4] As seen in the main slides, $z^*_{1-\alpha} = qnorm(1-\alpha)$.

Obtaining the p-value for One-Sided Tests

For the two-sided case, the p-value was found as follows:

```
2*(1 - pnorm(abs(z))) # 2-sided cases
```

For one-sided cases, if testing hypothesis tests with the alternative hypothesis $H_1 : \mu > \mu_0$:

```
1 - pnorm(z) # 1-sided case
```

And if testing hypothesis tests with the alternative hypothesis $H_1 : \mu < \mu_0$:

```
pnorm(z) # 1-sided case
```

Hypothesis Test, example revisited

Let's revisit the example from earlier with the more realistic hypothesis in play. The company that owns this server wants to claim that on any given day **more than** 150 users are connected to its server. Test this claim at the significance level of $\alpha = 0.05$.

- State hypotheses and choose level of significance.
- Collect data.
- Calculate summary statistics and p-value.
- Conclude.

Confidence Interval, example

Create a 95% **matching** confidence interval for the mean number of users connected to this server.

General Tip: Identify all your relevant variables and define them in R to help with your setup for the problem.

One-Sided Confidence Intervals, Bernoulli Case

When working with **proportions**, the confidence intervals won't extend to ∞, since p can only take values between 0 and 1. Instead, one-sided confidence intervals will be bound by these limitations in place of infinity.

If testing hypothesis tests with the alternative hypothesis $H_1 : p > p_0$, use an **upper** confidence interval:

$$[\hat{p} - z^*_{1-\alpha} \cdot \sigma_{\bar{X}}, 1].$$

If testing hypothesis tests with the alternative hypothesis $H_1 : p < p_0$, use a **lower** confidence interval:

$$[0, \hat{p} + z^*_{1-\alpha} \cdot \sigma_{\bar{X}}].$$

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Hypothesis Test, example

A trucking firm suspects that a spark plug manufacturer is lying about the average lifetime of its spark plugs. The manufacturer, Sparky, claims their spark plugs last $28,000$ miles on average. From a sample of 27 spark plugs, we find $\bar{X} = 26,040$ with standard deviation $\sigma = 12,894$. Test Sparky's claim about the average lifetime of their spark plugs.

- Establish hypotheses.
- Collect data.
- Analyze data.
- Make conclusion.

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Why Confidence Intervals?

The purpose of a confidence interval is to give a more informative estimator than a simple point estimator. You can achieve a certain level of confidence that your interval contains the true population mean, rather than hoping that your point estimator happens to be fairly close.

Remember, there are two parts to the margin of error in a two-sided confidence interval: **number of standard deviations** $z^*_{1-\alpha/2}$, and **size of the standard deviations** $\sigma_{\bar{X}}$.

Why Hypothesis Tests?

When you want to test a statement (H_0), a hypothesis test lets you compare your sample to the statement by determining how unlikely your test statistic is to occur under the assumption that statement is true.

We find the z-score for our test statistic $\bar{X}$ or $\hat{p}$, which represents how many standard deviations away from the mean the test statistic is.

By using pnorm() to find how unlikely such a test statistic is to occur, in relation to some level of confidence α, we can determine whether we reject or fail to reject the statement in question.

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